

Bureaucrats and the Choice of Institutional Priorities

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How do international organizations decide which issues to prioritize? Existing accounts emphasize how political principals determine institutional priorities. We shift focus to individual bureaucrats, theorizing that interactions between staff shape their choice of how to allocate resources across governance tasks. When bureaucrats interact with colleagues attentive to a given issue, social pressures and career concerns lead them to reallocate effort towards the governance of that issue, even when principals send mixed signals on its value. We test this argument in the case of climate change at the International Monetary Fund. With original data on professional networks within the IMF, we show that ties to climate-attuned colleagues increase the effort bureaucrats subsequently invest in climate relative to other policy areas, particularly when those colleagues are senior to them. These findings offer a social account of bureaucratic decision-making and show how intra-staff linkages can support institutional adaptability to emergent policy challenges.

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Formal mandates of international organizations (IOs) typically concern specific cooperation problems. The International Monetary Fund (IMF) is mandated to support international financial stability; the World Trade Organization is mandated to facilitate global trade flows; the United Nations is mandated to promote global peace and stability. Yet IOs often broaden their focus to encompass new policy challenges over time. International financial institutions, for instance, have “mainstreamed” issues of climate change and gender equity in recent years despite neither issue being explicitly tied to their founding charters (Clark and Zucker 2024; Heinzl, Weaver, and Jorgensen 2025). Such processes of mission creep often occur amid internal debate over the proper allocation of resources. One senior official at the IMF, for example, recently bemoaned the institution “getting so much into climate change [and] women in development ... new issues that are very important but outside of the IMF’s mandate.”¹

Political principals sometimes drive institutional reorientation to new issues (Nielson and Tierney 2003), but they often have divergent policy preferences (Copelovitch 2010a), including over which issues to prioritize. In practice, much of the day-to-day responsibility for distributing scarce resources across policy tasks lies with individual bureaucrats. Sociological institutionalists argue that policymaking slack enjoyed by bureaucrats produces separation between IOs’ legal mandates and their actual activities (Barnett and Finnemore 1999, 2004). However, the process by which international bureaucrats decide how to distribute effort across priorities — how they choose to spend their time and energy — is less well understood. Research documents static determinants of bureaucrats’ policy preferences, such as educational background and nationality (Chwiero 2015; Kaplan 2017; Nelson 2017; Clark and Dolan 2021), but these accounts are not suited to explain how the behavior of individual bureaucrats varies over time. Other work identifies how on-the-ground experiences of bureaucrats modify their awareness of different issues, but stops

¹Interview with former director of the European Department at the IMF (June 2021).

short of evaluating whether bureaucrats merely pay lip service to or invest meaningful effort in emergent issues (Clark and Zucker 2024). Amid debate among principals and staff on the proper direction of their institution, how do bureaucrats decide which policy tasks to prioritize?

We explore this question in reference to climate change, an issue that longstanding IOs have increasingly sought to address. We see climate as a novel and complex issue, making it a demanding test of how bureaucrats decide whether to invest effort in emergent policy domains. Climate has historically not been of interest to most IO bureaucrats, and significant effort is required to acquire expertise in the issue.² This is especially true within institutions established for purposes other than climate governance, such as the IMF and World Bank. The IMF, for example, traditionally employs neoliberal macroeconomists without formal training in climate-related issues (Chwieroth 2008), and powerful member states have disputed the value of incorporating climate into the Fund's operations (Clark and Zucker 2024). Indicative of a lack of preexisting climate expertise, the institution has sought to hire external climate specialists in recent years; a scarcity of such experts has complicated these efforts (Committeri et al. 2022).

We propose a social, intra-bureaucratic account of how IO staff decide whether to shift resources away from issues of traditional interest to climate. Bureaucrats at the IMF and other IOs do not work in isolation; they frequently operate in teams alongside colleagues. Examples abound: the Fund assembles mission teams with staff drawn from across departments to assess member states' economies and issue policy recommendations; the World Bank organizes task teams to manage development projects and consult with local stakeholders; the World Health Organization dispatches emergency teams during disease outbreaks. Because much of this work is organized through small, cross-departmental teams,

²Climate governance demands substantial inputs of specialized expertise (Gazmararian and Tingley 2023, 65).

bureaucrats routinely observe the preferences and priorities of their colleagues as they collect and interpret information, apply institutional rules and procedures, and deliver services.

We theorize that the experience of working alongside climate-attuned bureaucrats — those attentive to climate issues — leads colleagues to subsequently invest greater effort in climate policymaking, even where political principals send ambivalent or contradictory signals on its importance. Such intra-staff interactions modify the perceived costs and benefits of reallocating effort across issue areas through two channels. First, operating alongside staff attentive to climate augments the perceived career benefits of prioritizing climate over other issues, particularly when those staff are senior to a given bureaucrat. Second, these interactions socialize staff into placing greater non-material value on climate governance, inducing them to shift effort from other professional tasks out of a desire to fit in with colleagues or to emulate those they respect.³ Following these interactions, bureaucrats across an institution — including those initially inattentive to and without personal experience of climate change (Clark and Zucker 2024) — increasingly prioritize climate in their work, in turn producing more developed climate policy outputs and shifting the balance of their institution’s efforts across issue areas.

This account takes seriously the hierarchical structure of bureaucracies. Influential accounts of bureaucratic politics adopt bipartite models of bureaucracies, where an undifferentiated body of staff acts as the agent of political principals (e.g., Huber and Shipan 2002; Copelovitch 2010*b*); one influential account of bureaucratic decision-making explicitly “neglect[s] the internal organization of different policymaking institutions” (Alesina and Tabellini 2008, 427). Other work focuses on bureaucrats of different ranks in isolation (Heinzel, Weaver, and Jorgensen 2025). These approaches are valuable, but they bracket questions of how interactions within and across ranks differentially affect staff behavior.

³This argument aligns with arguments that “bureaucrats’ attitudes [about climate policy] are influenced by the degree to which they perceive the wider environment to support and legitimize their ideas” (Dellmuth 2023, 1).

We theorize that the effects of intra-staff interactions are asymmetric across ranks. We specifically propose that bureaucrats are most influenced by colleagues senior to them.⁴ While bureaucrats may be socialized by junior colleagues and staff of equal rank, they should become especially confident in both the career-related benefits and the appropriateness of expending effort on climate upon engagement with climate-attuned senior colleagues. They wield greater influence over promotions and assignments, shaping the perceived career returns to reallocating effort, and they also serve as respected reference points for appropriate professional conduct. Career and socialization pressures should consequently be strongest when bureaucrats work alongside climate-attuned senior colleagues, leading preferences around issue prioritization to flow down the bureaucratic ranks.

To test this theory, we introduce original data on the professional interactions of thousands of individual IMF staff, as well as the content of over 11,000 policy recommendations issued by the IMF to member states between 2010–2019. Bureaucrats craft these policy recommendations under tight time constraints, forcing them to choose how to allocate their time across different domains. To measure how bureaucrats distribute effort across issues, we compare the linguistic complexity and specificity of recommendations — two measures of analytical depth, which we suggest reflects greater investment of effort by bureaucrats — issued by staff in different policy domains. We show that the detail and sophistication of recommendations increases with linguistic complexity, supporting its use as a proxy for the amount of effort invested by staff (see Wasow 2026). We complement these analyses with evidence drawn from interviews with past and present Fund staff, as well as tests of semantic similarity between bureaucrats’ climate recommendations.

We find robust support for our theory. To identify how individual staff reallocate effort, we measure changes in the complexity of climate-related policy recommendations to which

⁴By senior colleagues, we refer to higher-ranking bureaucrats (senior career civil servants), not institutional executives or political principals.

a given bureaucrat contributes, in relation to the contemporaneous complexity of their non-climate recommendations. Analyses show that staff produce relatively more complex climate recommendations in the years *after* working alongside climate-attuned colleagues. This is particularly so when those colleagues are senior to them, consistent with the idea that top-down cues are especially salient when bureaucrats decide where to invest their time and energy. Bureaucrats who lack such climate-attuned professional networks issue climate policy recommendations that are relatively less sophisticated. We focus on recommendations issued in years after interactions with climate-attuned colleagues to capture whether reformed preferences are carried into the future, and to avoid conflating changes in bureaucratic effort with the contemporaneous presence of climate-attuned colleagues. Evidence suggests that these patterns are unlikely to be artifacts of biased selection into exposure to climate-attuned colleagues.

This paper makes several contributions. First, it illustrates how bureaucrats navigate contexts of multiple and conflicting demands on their labor. Bureaucrats receive inconsistent signals from political principals on the importance of climate and observe variation in colleagues' embrace of the issue. The limited amount of effort individual bureaucrats can afford to invest in their work forces them to make tradeoffs between climate and issues of longstanding interest to their institution; in other words, bureaucrats must decide which demands to act on and which to deprioritize. How they choose to make this tradeoff is consequential. In technocratic settings, individuals face incentives to both exhibit high degrees of technical precision across policy tasks and signal attentiveness to relevant issues; perceptions of misallocated effort or misplaced priorities impose substantial costs on bureaucrats' career trajectories. Our account illustrates how bureaucratic hierarchies guide individuals' decision-making under these conditions. Individual staff are responsive to the climate attitudes of senior bureaucrats, and weakly or unresponsive to colleagues who have less sway over career trajectories and exert milder socialization pressures.

The paper moreover offers a new conceptualization of bureaucratic decision-making that centers the role of internal staff interactions. Scholars of bureaucracy often focus on time-invariant determinants of bureaucratic behavior, including elite educational backgrounds (Weaver 2008; Chwiero 2015; Kaplan 2017) and other biographical characteristics such as prior work experience (see Krcmaric, Nelson, and Roberts 2020 for a review). This paper considers how bureaucrats' allocation of effort across tasks varies over time as a function of interpersonal interactions, rather than being fixed by background characteristics or initial training.⁵ Further, work on domestic and international bureaucracies typically focuses on how staff interact with counterparts *external* to their institution, such as legislators (Huber and Shipan 2002), interest groups (Thrall 2025), foreign officials (Malis 2021), and staff at other international institutions (Sommerer and Tallberg 2019). Less attention has been paid to how intra-bureaucratic networks shape staff behavior and performance. Recent work focuses on how individual staff learn in isolation, such as from their personal experiences while on assignment overseas (Clark and Zucker 2024). This paper illuminates how patterns of bureaucrat behavior diffuse across an institution via intra-staff engagement.

In doing so, this paper builds on a growing body of work that emphasizes preference heterogeneity within international (Heinzel, Weaver, and Jorgensen 2025; Lang, Wellner, and Kentikelenis 2025) and domestic bureaucracies (Schub 2022; Jost 2023; Carcelli 2024). While this literature has identified how such heterogeneity can initially emerge, it has less to say about how it evolves: whether bureaucracies remain divided attitudinally, converge to common understandings of given issues, or resolve to some other distribution of preferences. This paper sheds light on this process, suggesting that regular interactions between staff — in particular, the administrative practice of assigning bureaucrats to work on small teams — serve as conveyor belts for disseminating attitudes and shifting decision-making

⁵An important literature considers socialization within IOs, though often focuses on national representatives more than career bureaucrats (e.g., Johnston 2008; Broome and Seabrooke 2015). Hooghe 2005 finds limited evidence of IO socialization of individual staff.

across large, hierarchical bureaucracies. These claims align with theories depicting political institutions as social environments that modify actors' preferences (Katzenstein 1996; Gheciu 2005; Adler 2008; Johnston 2008). We extend this work by illustrating the social process by which bureaucrats allocate effort to emergent and controversial issues.

Our findings further speak to debates around institutional adaptability. Important scholarship portrays institutions as rigid and resistant to change, with powerful external shocks needed to shift their trajectories (Krasner 1976; Wallander 2000; Colgan, Keohane, and Van de Graaf 2012).⁶ Contributing to institutional inflexibility is homogeneity in bureaucratic preferences. When the staff of institutions share similar worldviews, there is little space for new ideas to take hold (Weaver 2008). Institutions reinforce this homogeneity by hiring staffers with shared educational and demographic characteristics (Chwioroth 2015; Nelson 2017) and by strictly monitoring bureaucrat behavior (Honig 2019). These literatures thus present change as rare, limited, and inhibited by individuals within the institution. In contrast, we identify an underappreciated, endogenous source of institutional evolution: interactions between bureaucrats, facilitated by routine personnel management practices that structure who works with whom and shape whose priorities are most visible. This interpersonal process is a means by which institutions may internally adapt to emergent policy challenges.

Climate change serves as our empirical focus, but we expect similar processes to operate in other cases. Artificial intelligence, cybersecurity, and digital finance are of growing interest to long-standing political institutions, yet are distant from many organizations' legal mandates and day-to-day operations. As with climate, institutions often lack deep in-house capacity in these domains, and surging demand for specialists has complicated efforts to rely on hiring of external specialists.⁷ We highlight a generalizable mechanism

⁶Incremental reforms can be pursued absent such shocks (Blyth 2002; Lipsky 2015; Carnegie and Clark 2023).

⁷See, e.g., U.S. Government Accountability Office 2025 [perma.cc/FL3L-NCZ7].

by which staff invest in the governance of emergent issues, and may ultimately improve the work of their institutions in those domains, even in the absence of unified principal directives or large-scale organizational restructuring.

SOCIAL THEORY OF BUREAUCRATIC DECISION-MAKING

Bureaucrats perform policymaking and implementation tasks that often span multiple issue areas (Barnett and Finnemore 1999; Jupille, Mattli, and Snidal 2013; Clark and Zucker 2024). At development organizations, staff manage portfolios that range from poverty alleviation to corruption to gender equity. At institutions of financial governance, bureaucrats evaluate diverse factors affecting macroeconomic stability, including sovereign debt, exchange rate regimes, trade exposure, labor market dynamics, and geopolitics. In such settings, no single bureaucrat can devote full attention to all relevant issues. Given limits to how much time and effort bureaucrats can devote to their work, individual staff must choose which issues to prioritize — which issues to research, to speak with policymakers about, to collect data on, to design and advocate policies for — and which to deemphasize.

Investing greater effort in an issue allows bureaucrats to do better work in that space. It permits more precise policy analyses, more appropriate and credible recommendations for member states, and more effective policy implementation. The tradeoff, however, is that effort reallocated to one issue diminishes the quality of work on other issues. This tradeoff is especially acute when bureaucrats confront novel issues that require breaking from existing modes of analysis or acquiring unfamiliar forms of knowledge. Insofar as technocratic bureaucracies expect and reward technical precision, staff may hesitate to address emergent issues without having previously invested significant effort in obtaining some degree of expertise in them.

Scholarship often explains how bureaucrats navigate this tradeoff via the lens of dele-

gation and incentive alignment. In these accounts, bureaucrats allocate effort in response to signals and material incentives issued by political principals (Gailmard and Patty 2007), adjusting their behavior to match institutions' formal mandates and principals' priorities (Pollack 1997; Clark and Dolan 2021). Such arguments are powerful, but they presume that principals issue relatively clear and consistent signals about which tasks will be evaluated and rewarded (cf. Copelovitch 2010a).

In practice, bureaucrats frequently operate under conditions of uncertainty. Stakeholders may disagree about organizational priorities, particularly when institutions expand into new policy domains like climate change (Clark and Zucker 2024; Clift 2025). Formal mandates may lag behind evolving expectations — a product of institutional rigidity and path dependence (Kaya 2015) — leaving bureaucrats unsure about the professional returns to reallocating effort to new areas. While principals can intervene in IO policymaking, they often refrain from doing so regularly, granting staff leeway in carrying out the day-to-day work of the institution (Stone 2008). Under these conditions, bureaucrats must infer which issues are safe, appropriate, and advantageous to prioritize.

A long tradition in international relations conceives of IOs not merely as instruments of powerful principals, but as social environments in which knowledge, norms, and professional priorities are produced and diffused. Research on epistemic communities highlights how shared causal beliefs and professional standards shape how policymakers interpret problems and evaluate solutions (Haas 1992). Sociological approaches similarly emphasize how repeated interaction within institutions generates convergence in preferences, identities, and understandings of appropriate conduct (Katzenstein 1996; Finnemore and Sikkink 1998). From this perspective, bureaucrats' decisions about where to invest effort are shaped not only by formal mandates, but also by their position in professional communities of practice (Hooghe 2005; Adler and Pouliot 2011). This logic is central to work on socialization within international institutions, which demonstrates how sustained

professional interaction drives acceptance of norms and ideas (Gheciu 2005; Adler 2008; Johnston 2008). Recent work applies similar insights to IO bureaucracies, documenting “contagion effects” through which attention to emergent issues, such as gender, diffuses across staff (Meier and McCrea 2022; Heinzl, Weaver, and Jorgensen 2025).⁸

We build on these literatures by theorizing that peer interaction helps bureaucrats resolve uncertainty about effort allocation across issue areas. Rather than treating expertise or attention as fixed attributes, we emphasize micro-level processes through which bureaucrats update beliefs about which issues are valued, legitimate, and professionally consequential. In doing so, we link macro-level claims about epistemic authority and norm diffusion to the everyday team-based interactions through which international organizations govern global issues — building on theories of bureaucratic decision-making that abstract away from the internal organizational structures of political institutions (e.g., Alesina and Tabellini 2008).

Peer Interaction and Effort Allocation

We theorize that interactions between a bureaucrat and climate-attuned colleagues, those who see climate as of interest and relevance to the institution,⁹ increase the effort subsequently invested by that bureaucrat in climate-related issues. They do so in two complementary ways.

First, interactions with climate-attuned colleagues provide information about the professional returns to climate work. Bureaucrats are constrained in the total amount of effort that they can devote to their work. Investing effort in a novel issue requires diverting attention from established tasks, which may carry career risks when organizational priorities are

⁸Experimental work similarly shows that group settings shape how policymakers evaluate information and navigate uncertainty, in part by revealing colleagues’ preferences and exerting social pressure to conform (Wayne et al. 2025). Research in educational psychology likewise highlights the benefits of cooperative learning in small groups (Slavin 2013).

⁹We do not theorize about the original source of institutional attention to climate. Prior work suggests this owes to the personal exposure of staff to climate impacts (Clark and Zucker 2024) or shifts in principal and executive preferences (Skovgaard 2021).

unsettled (i.e., when principals express uneven preferences). Exposure to colleagues who visibly engage with climate reveals information about the distribution of attitudes within the institution and the extent to which climate work is tolerated or rewarded. Such interactions can reduce perceived risk by signaling that climate engagement is professionally viable, correcting common underestimates of the extent to which others support climate action (Mildenberger and Tingley 2019). Updated beliefs about the extent of pro-climate attitudes in one's institution, and level of climate knowledge among colleagues, may also exacerbate the perceived risk of *not* attending to climate — especially in meritocratic settings where lagging behind peers can have reputational costs (Zhou 2005; Sauder, Lynn, and Podolny 2012).

Second, peer-to-peer interactions socialize bureaucrats into valuing climate work. Entrepreneurial bureaucrats may actively persuade colleagues of the relevance of climate to their institution's mission, leading staff to shift effort from other issue areas to climate. Such a “rational learning” model implies bureaucrats treat their peers as sources of information and adapt based on perceived policy effectiveness or technical merit (Meseguer 2006; Gilardi 2010). Subtler socialization dynamics may also emerge. Johnston (2008) documents attempts by individuals at international organizations to “fit in” with colleagues by changing their behavior. Johnston identifies patterns of mimicry, where an actor “sees others anticipating benefits [from some action] and assume that it too can hope to gain.” While the motive here may be material, Johnston notes that mimicry is also appealing insofar as it helps individuals avoid being “viewed by others as out of fashion, behind the times, and thus missing out on a status-enhancing experience” (46). In this sense, socialization does not merely transmit information; it shapes perceptions of appropriateness and professional identity.

Together, these mechanisms imply:

Hypothesis 1. *Bureaucrats who work alongside climate-attuned colleagues should subsequently invest greater effort in climate.*

Hierarchy and Asymmetric Socialization

Bureaucracies, both international and domestic, are hierarchical. Hierarchy conditions how social cues are interpreted and how uncertainty is resolved. We propose that senior colleagues differ from other staff in two respects: they wield evaluative authority over career trajectories (i.e., promotion decisions) and occupy positions of symbolic status within the organization. As a result, staff should be especially responsive to interactions with climate-attuned senior colleagues.

From a career perspective, senior colleagues exert influence over promotions, assignments, and reputational standing. Their own interest in climate, and belief of its relevance to the broader institution, should suggest that they would reward staff for investing some effort in climate. While superiors within the bureaucracy will not necessarily provide more or better information about climate than colleagues of equal or lower rank, their engagement with climate signals that reallocating effort toward the issue is professionally acceptable and potentially rewarded, alleviating uncertainty about the returns to such investment.

From a socialization perspective, senior colleagues serve as respected reference points for appropriate professional conduct. Organizational research suggests that individuals are particularly likely to emulate behaviors modeled by successful or authoritative figures, especially under conditions of uncertainty (Finnemore and Sikkink 1998; Dobbin, Simmons, and Garrett 2007). Observing senior colleagues prioritizing climate thus provides a powerful cue about how the institution defines valued work. While social influence may also operate horizontally among peers, and occasionally upward from junior staff, we expect career-motivated bureaucrats to look upward when evaluating which issues merit sustained attention. Hierarchy therefore amplifies both informational and socialization mechanisms,

producing asymmetric effects across ranks. This leads to our second hypothesis:

Hypothesis 2. *In deciding how to allocate effort to climate, bureaucrats should respond more to climate-attuned senior colleagues than those of equal or lower rank.*

CASE OF THE IMF

We test this theory in the case of the IMF, though see it as applicable across hierarchical and technocratic institutions where staff retain some discretion over how to allocate resources across issue areas. The IMF is an appealing empirical case as its staff lack clear top-down direction on the value of prioritizing climate change. While Fund staff have grown more attentive to the issue in recent years, member states and officials disagree on its importance relative to the macroeconomic risks of traditional interest to the institution. The United States, the IMF's largest shareholder, has accused the Fund of "devot[ing] disproportionate time and resources to work on climate change" at the expense of "its work on critical macroeconomic issues."¹⁰ Other states and stakeholders have conversely lobbied the IMF to incorporate climate more into its operations, exemplified by the Bridgetown Initiative effort to "reshap[e] the [international] financing system to ... spur climate action."¹¹ Fund management has both emphasized climate change as a threat to "macroeconomic and financial stability" and occasionally questioned the extent to which staff should prioritize the issue.¹² Issues of climate governance are moreover marked by "debate" and "dissonance amongst [Fund] staff" (Clift 2025, 1273).

For individual staff within the Fund, the question of how to allocate time and other resources across climate and non-climate issues is thus a salient one. Accentuating this tradeoff is a perception among staff that climate work demands significant investments

¹⁰U.S. Department of the Treasury 2025 [perma.cc/J23J-5RRX].

¹¹Bridgetown Initiative 2025 [perma.cc/MCS6-BTN4].

¹²X 2024 [perma.cc/HTA5-WAJC]; Reuters 2025 [perma.cc/KJX8-BAL6].

of effort. In interviews, IMF officials expressed that, historically, many staff did not see their skills in macroeconomic analysis as easily applicable to climate. Analytical work on climate finance, for instance, proved “very difficult” as staff were initially wary of whether they “were equipped to work in that space,” and believed that significant effort would be required for them to do good work.¹³ While some staff chose to remain focused on standard macroeconomic issues, such as debt and balance of payments issues, others have invested more time in learning about climate and incorporating it into their work.¹⁴

To evaluate whether our theory explains how bureaucrats navigate these tradeoffs, we collect original data on the career paths and professional networks of individual IMF staff, as well as the sophistication of IMF policy recommendations, which we assume increases with the effort expended by contributing bureaucrats. We draw these data from Article IV reports, which are the products of routine surveillance missions undertaken by IMF staff teams to individual member states. Missions “primarily [involve] meetings with senior and midlevel officials from the finance ministry, central bank, and other government ministries involved in the formulation of economic policy.”¹⁵ The reports, authored by small groups of IMF staff, detail the state of countries’ economies, identify macroeconomic risks, and offer policy advice to host country governments.¹⁶ They exemplify the IMF’s role as a provider of detailed and influential economic policy advice (Breen and Doak 2023; Cormier and Manger 2022). Article IV teams consist of staff recruited from both functional departments (e.g., Monetary and Capital Markets) and area departments (e.g., Asia and Pacific).¹⁷

Bureaucrats have meaningful discretion in the construction of Article IV reports, mak-

¹³Interview with official in the IMF Strategy and Policy Review Department (October 2025).

¹⁴Interviews with officials in the IMF Strategy and Policy Review Department (October 2025) and Research Department (January 2026).

¹⁵IMF 1999 [perma.cc/ZGL4-5QQC], p. 25.

¹⁶Because multiple staff contribute to these reports, they are noisy, but still meaningful (Clark and Zucker 2024), indicators of the preferences and behaviors of individual contributing bureaucrats.

¹⁷Countries are surveyed every one and a half years on average in our data set. Political instability can lead to delays (IMF 2018, perma.cc/AA2U-SPBL).

ing them an appropriate place to document the effects of bureaucratic effort. These written outputs are leading indicators of policy shifts within IOs (Cormier and Manger 2022), and can affect the behavior of target governments (Goes and Chapman 2024) and market actors (Breen and Doak 2023). Bureaucrats have been shown to impact even higher-stakes outputs subject to more member state scrutiny, such as the design of IMF conditionality programs (Lang, Wellner, and Kentikelenis 2025).

We collected data on 581 Article IV reports issued between 2010–2019, a period during which IMF attention to climate grew dramatically.¹⁸ Based on the listed authors of these reports, we identify 879 IMF personnel — resident representatives, mission chiefs, and rank-and-file mission team members¹⁹ — who contributed to multiple reports during these years. Using the reports that each staff member contributed to, we record the countries that each individual worked in, when they worked there, and, critically, the other IMF personnel on the mission teams they were assigned to. We link bureaucrats across reports on the basis of author lists included on report cover pages, allowing for small differences in name spelling across reports (see Appendix A for matching procedure details). Our data include 2,239 unique bureaucrat-mission observations.

Policy Recommendations

To evaluate bureaucrats’ investment of effort in climate-related policy tasks, we introduce new data on individual policy recommendations issued by the IMF to member states. Our research team manually extracted 11,608 unique policy recommendations from these 581 Article IV reports. These policy recommendations are primarily delivered in the narrative

¹⁸Note that this period precedes a 2021 reform that “commit[ted] the IMF to adopt a systematic approach to integrating climate-related risks” into its surveillance activities (Kentikelenis and Stubbs 2025, 38).

¹⁹Resident representatives are IMF staff based in an assigned member state. Mission chiefs are typically drawn from IMF area departments and lead the Article IV missions. Rank-and-file staff are drawn from area departments and functional departments. Mission teams include one resident representative, one mission chief, and fewer than ten rank-and-file staff.

section of the reports, with short passages describing economic challenges and advising on proper policy directions. To identify climate-related recommendations, we conducted keyword searches on this extracted text based on a dictionary of climate- and decarbonization-related terms, which we then verified using an automated large language model (LLM)-based classifier. We identify 234 recommendations that explicitly concern climate change or decarbonization, spread across 24% of all sampled reports; 87% of these were likewise coded as climate-relevant via the LLM classifier. Appendices B, C, and D detail the recommendation extraction, keyword search, and LLM validation processes.

In developing policy recommendations, staff must choose which issues to prioritize. The surveillance missions that culminate in Article IV reports occur on short timelines, typically less than two weeks in length. On such missions, staff decide whom in a member state to consult with, what topics to investigate, and what data to compile. IMF guidance for its staff notes that “selectivity is critical” in how they conduct their missions; staff are “not expected to cover all issues ... and should avoid a check-list approach.”²⁰ Staff additionally must choose how to allocate effort across a range of possible policy recommendations. Composing climate recommendations can be especially time intensive. Development of “carbon pricing [recommendations] is highly complex,” reported Fund staff in interviews. “It requires a lot of interdisciplinary research to combine the scientific models, the data analytics, the financial economics, and the fiscal economics to form policy lessons for countries.”²¹

The share of climate-related recommendations, as well as staff exposure to climate-attuned coworkers, has grown over time. As illustrated in the left-hand panel of Figure 1, climate accounted for just 0.4% of all recommendations in 2010; by 2019, this share had increased nearly sevenfold to 2.7%. Indicative of diffusion in climate interest across the

²⁰IMF 2022 [perma.cc/FY54-QW5Y], p. 7.

²¹Interview with officials in the IMF Monetary and Capital Markets Department (July 2021).

Fund, we find rapid growth in staff members’ average exposure to climate-attuned colleagues over time (right-hand panel of Figure 1). In 2015, just 11% of bureaucrats in our data had worked alongside a climate-attuned colleague (defined here as someone who previously issued at least one climate-relevant recommendation) on an Article IV mission. By 2019, that share had quintupled to 58%. In this paper, we bracket the question of what drives initial interest in climate within an institution; prior work attributes this to on-the-ground learning by staff (Clark and Zucker 2024) and the initiatives of high-level executives (Skovgaard 2021).

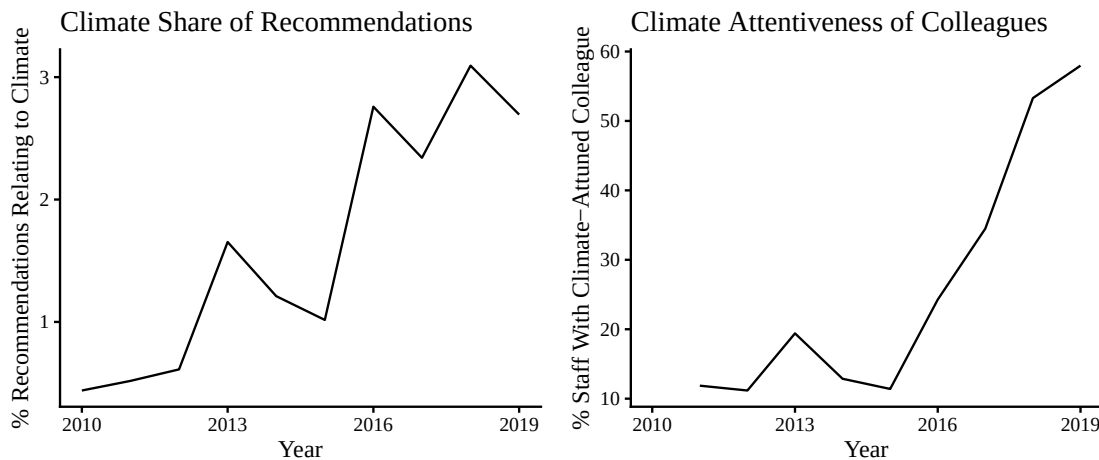


Figure 1: The left-hand panel depicts the share of policy recommendations each year that concern climate change. The right-hand panel depicts the percentage of IMF bureaucrats who previously worked with at least one climate-attuned colleague on an Article IV mission.

We first describe the content of climate policy recommendations. Of interest is how the Fund balances attention to adaptation (policies intended to bolster economic resilience to physical climate disruptions) and mitigation (efforts to reduce carbon emissions). Using a keyword-assisted topic model (Eshima, Imai, and Sasaki 2024), we find that adaptation-related recommendations exceed mitigation-related recommendations in frequency by 33 percentage points (see Appendix E). IMF staff likely place greater emphasis on climate adaptation than on mitigation because adaptation policies directly address the near-term

macroeconomic risks posed by climate change. Extreme weather events and climate volatility generate balance-of-payments pressures, disrupt growth, and erode tax revenue and governments' ability to service debt, all of which are core concerns of the Fund's surveillance mandate. Adaptation measures that reduce vulnerability to climate shocks therefore align with IMF staff's existing analytical frameworks and responsibilities.

We further characterize the types of policies that staff emphasize in discussing climate. To do so, we instruct an LLM to categorize the policies in each recommendation (with manual verification) as: natural disaster management, including general adaptation policies; fiscal policy, including policies concerning debt management, deficits, and fiscal buffers; public sector employment and bureaucracies; insurance; tax policy, including carbon prices; labor policy; infrastructure investments; ease of doing business; monetary policy; welfare policies; financial sector risk; and other mitigation and decarbonization policies.²² Appendix F provides further detail on this coding procedure.

Figure 2 depicts the categories of policies discussed in climate-related recommendations, as well as overlap across pairs of policy categories. Mirroring the aforementioned emphasis on adaptation, the majority of recommendations concern policies focused on natural disaster management and other forms of adaptation. Many of these disaster-focused recommendations include discussions of infrastructure investment, typically to harden infrastructure against climate shocks, and fiscal policy, often with an eye to building “fiscal buffers” to absorb the costs of climate disruptions.²³ In 2018, for example, staff advised that the Solomon Islands target “an overall deficit of 1.5 percent of GDP, which would balance containing debt levels with addressing infrastructure needs and consistent with building a fiscal buffer for natural disasters.” To manage “rising temperatures and sea level[s]” in Tuvalu, staff recommended in 2016 that the country's macroeconomic framework adopt “a

²²These categories were chosen via a manual review of a sample of climate-related policy recommendations.

²³This is consistent with the findings of Kentikelenis and Stubbs 2025.

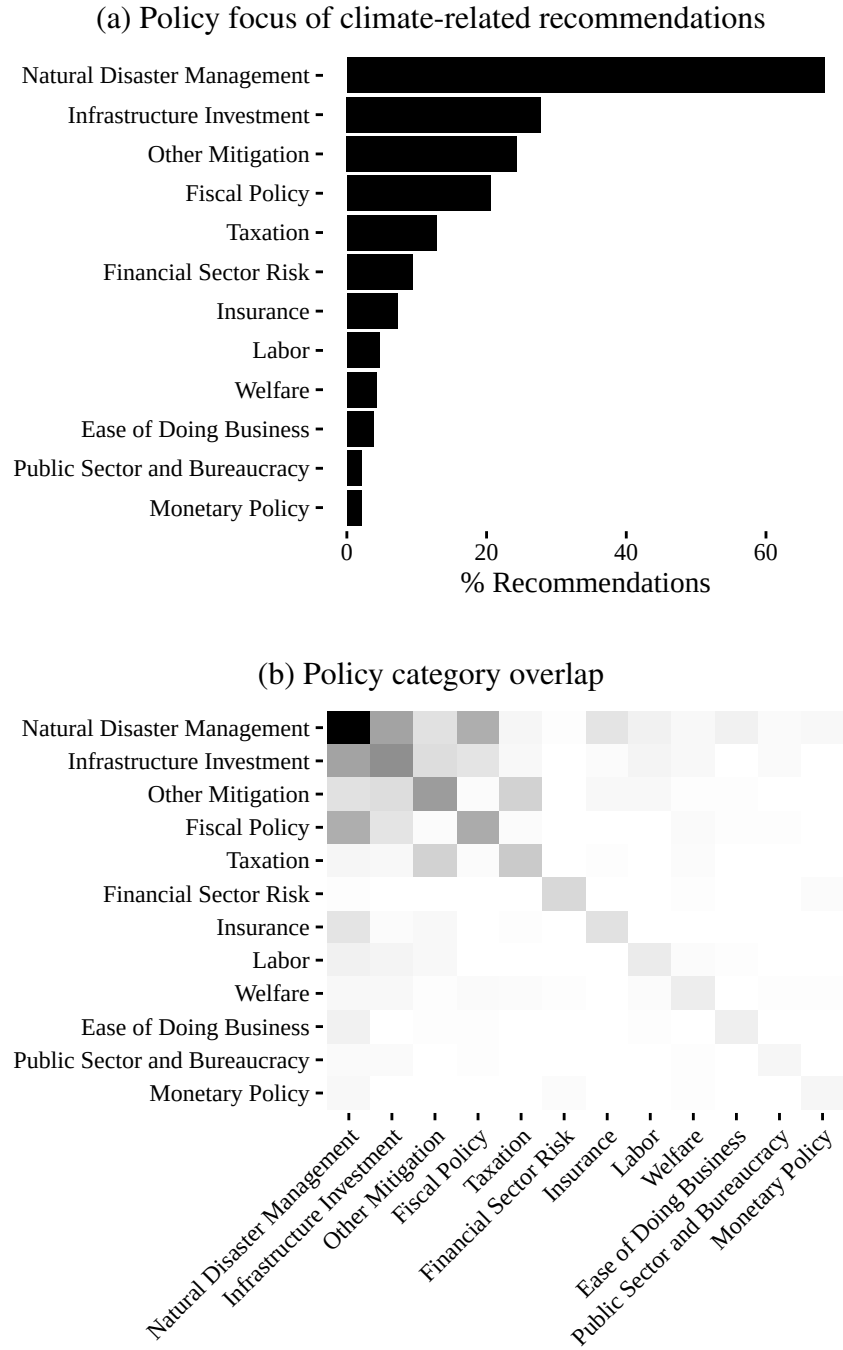


Figure 2: Panel (a) depicts the proportion of climate-related recommendations concerning each of twelve policy categories, as classified by the GPT-4o large language model. Note that individual recommendations may be sorted into multiple categories. Panel (b) depicts a matrix of policy categories, shaded by the count of recommendations categorized into each policy pair. Cells along the diagonal indicate recommendations sorted into a single policy category.

forward-looking, multi-pillar approach: (i) risk assessment and risk reduction, (ii) sustaining fiscal buffers, and (iii) international risk sharing.”

Recommendations concerning climate change mitigation often involve discussions of carbon prices, including carbon taxes, and renewable energy. In 2018, for instance, staff recommended that Vietnam “lower the intensity of dirty fossil fuels by raising the contribution of renewables and gas; provide stronger incentives for green growth, including full carbon pricing of fossil fuels while investing in infrastructure more resilient to natural disasters; and introduce an arms-length regulatory framework to prevent and manage environmental risk.” To China in 2016, staff advised that “substantially raising taxes on fossil fuel and pollution (e.g., a carbon or coal tax) would help curtail emissions, improve energy efficiency, raise revenue and prevent almost 4 million premature deaths by 2030.”

Measurement of Effort

Measuring the issue-specific and time-variant effort of individual bureaucrats is challenging. Here, we leverage the text of individual policy recommendations to estimate the effort invested by contributing authors, building on recent methodological work that treats the depth of written text as indicative of “effortful action” (Wasow 2026, 1). Our approach assumes that bureaucrats who invest greater effort in a given issue area will write more *complex* policy advice in that area: recommendations with greater analytical depth, detail, and specificity to a given country context.²⁴

We measure the complexity of policy advice in two ways. First, we calculate the diversity of word choice in recommendation text. We do so by computing the lexical richness of the text of each recommendation: “the number of different terms used in a text and the diversity of the vocabulary” (Torruella and Capsada 2013, 448). Lexical richness is com-

²⁴As noted previously, recommendations are composed by small groups of staff. If prior exposure to climate-attuned colleagues shapes individual behavior, bureaucrats with such exposure should elevate the complexity of the recommendations they subsequently coauthor.

puted via Carroll’s corrected type-token ratio, defined as $V/\sqrt{2N}$, where V is the number of “types” (unique words) present in a string and N is the count of words in that string. Lexical richness is an appropriate measure because recommendations that are more detailed require greater analytical investment by authors. Recommendations that draw on a wider and more varied vocabulary tend to reflect more extensive engagement with country conditions, consideration of multiple policy instruments, and the integration of technical concepts. By contrast, low-effort recommendations are often brief, generic, and can be produced with minimal issue-specific analysis, indicating less investment of effort.

Recommendation	z-score
In light of the authorities’ priority investment needs in infrastructure and electricity, as well as climate change mitigation and adaptation investments over the long term, the share of capital spending should increase in the medium to long term. Keeping a firm grip on current spending will be essential.	−0.9
The Bahamas has traditionally absorbed frequent natural disaster shocks ex post through the public balance sheet. An increased reliance on ex ante mitigation policies would help reduce and smooth the economic and fiscal impact of natural disasters. To this end, staff recommended integrating a well-designed natural disasters savings arrangement into the fiscal framework, insuring public assets through private natural disaster insurance and incentivizing the population to use these instruments more widely, including through targeted subsidies to improve affordability for low-income households - and making sure that building regulation, land use, and zoning guidelines are adequate and are reviewed and updated frequently.	1.6

Table 1: Sample of IMF policy recommendations with z-scores of Carroll’s corrected type-token ratios (measure of complexity).

Table 1 lists example recommendations at low and high levels of complexity. As the examples suggest, low-complexity recommendations are often short and involve fairly non-specific advice. More complex recommendations tend to be longer, exhibit deeper engagement with local conditions in a given country, and offer more specific and actionable policy advice. These text features suggest higher levels of effort invested by contributing authors

(Wasow 2026), in terms of both doing research on a given issue and composing the ultimate recommendation. Insofar as bureaucrats pay costs for issuing low-quality advice, which we see as plausible given the IMF’s emphasis on technical rigor,²⁵ they are unlikely to casually incorporate detail and specificity into their analyses.

Second, we develop an index to capture the specificity of language in policy recommendations. The index measures the extent of quantitative detail provided in reports (inclusion of numerical digits, percentages, and monetary figures), references to explicit policy instruments, and incorporation of time-horizon markers (e.g., short, medium, or long term; “by x year”). We likewise assume that more specific recommendations reflect greater investments of effort by contributing bureaucrats. Appendix G details construction of this specificity index and plots its correlation with the preceding metric of lexical richness.

ALLOCATION OF BUREAUCRATIC EFFORT

We now test, per our first hypothesis, whether bureaucrats who interact with climate-attuned colleagues subsequently invest greater effort in climate policy recommendations. To do so, we estimate the following model by ordinary least squares. This model is estimated at the bureaucrat-report-policy category level; for each bureaucrat in our sample, we have two observations for each report they authored, one the aggregate of climate policy recommendations (excluded if none) and the other the aggregate of non-climate recommendations.²⁶

$$\text{complexity}_{ictk} = \beta \left[\mathbb{I}(\text{climate}_k = 1) \times \text{prior coworker attention}_{i(t-1)} \right] + \gamma \mathbf{Z}_{ict} + \delta_i + \zeta_{(c)t} + \varepsilon_{ictk}$$

²⁵IMF 2022 [perma.cc/FY54-QW5Y].

²⁶As illustration, suppose a bureaucrat Smith authored a report for India in 2019. We would have separate observations for Smith-India-2019-climate (aggregate of climate-related recommendations) and Smith-India-2019-nonclimate (aggregate of other recommendations in the same report).

The model regresses the mean complexity of policy recommendations coauthored by bureaucrat i in a report for country c in year t on the climate relevance k of that recommendation,²⁷ interacted with the climate attentiveness of colleagues with whom the bureaucrat worked on *previous* Article IV missions. We calculate colleagues’ climate attentiveness as the count of climate-related policy recommendations they issued prior to working alongside bureaucrat i on a mission. For each bureaucrat i , we then take a rolling sum of colleagues’ climate attentiveness through year $t - 1$ (log transformed). We choose a count measure of exposure to climate-attuned colleagues because we do not conceive of this as a one-time shock; exposure to climate-attuned colleagues should have an additive effect on subsequent investments of effort in climate, and should compound when those colleagues previously worked on climate to a greater extent. Thirty-seven percent of bureaucrats in our sample had at least one interaction with a climate-attuned colleague by any given year.

We control for the constituent terms of the interaction, as well as for a bureaucrat’s climate attentiveness (count of prior climate recommendations). The latter helps account for the possibility that colleague exposure and recommendation complexity are confounded by an individual’s preexisting climate attitudes. We further control for the number of years that bureaucrat i has been in the sample to avoid conflating coworker attention with linear time trends. Additional covariates include the count of climate disasters experienced in country c , as well as the share of other current mission team members who previously issued climate-related policy recommendations.²⁸ Both of these controls account for contemporaneous drivers of climate attention (see, e.g., Clark and Zucker 2024). Bureaucrat, year, and report (country-year) fixed effects absorb unobserved heterogeneities across individuals, time periods, and mission teams.²⁹ We cluster standard errors by bureaucrat. Our

²⁷Results are consistent when redefining the outcome as median complexity.

²⁸To measure climate disasters, we aggregate climatological and meteorological disasters recorded in the EM-DAT database.

²⁹As country-year fixed effects absorb year fixed effects, we include these in separate models.

primary estimand of interest is β , the effect of prior coworker attentiveness to climate on the complexity of climate-relevant recommendations, relative to non-climate recommendations.

Inference

This model, which compares climate to non-climate recommendations, is useful for at least two reasons. First, it allows us to account for level differences in the typical complexity of recommendations that concern climate versus other issues. Due to the unique vocabulary of climate change, discussions of the issue may be recorded as more complex than discussions of other issues where, for unobserved reasons, the language used tends to be less lexically rich. The model captures changes in the complexity gap between climate and non-climate recommendations, which we assume owes to changes in the allocation of bureaucratic effort between climate and other issues.

Second, this design addresses the possibility that the total effort invested by bureaucrats varies across reports. For unobserved reasons, bureaucrats may invest more effort in certain reports, increasing the complexity of all recommendations included therein. Examining climate-related recommendations in isolation would mistake this increase in complexity for a reallocation of effort to climate, rather than an increase in effort across issue areas. This approach, by capturing the complexity of climate discussions relative to other issues, better reflects our interest in how bureaucrats redistribute effort across tasks.

The identifying assumption behind these tests is that staff exposure to climate-attuned colleagues is independent of their prior attitudes towards climate change. Readers may be concerned that staff do not randomly vary in their exposure to climate-attuned colleagues. It is possible, for instance, that managers who themselves are interested in climate deliberately assemble teams of climate-attuned staff and encourage them to devote special attention to climate-related policy recommendations. This would confound the relationship

between colleagues' climate attentiveness and the contemporaneous complexity of climate outputs. Here we reiterate that our analyses focus on how team assignments affect *future* investments of effort in climate; in other words, how staff behave after being rotated off a given team. If bureaucrats are solely responsive to managerial directives, we should not expect engagement with climate-attuned colleagues in one period to be associated with an increased climate focus in a future period. We further reiterate that we include person fixed effects, which absorb time-invariant differences in bureaucrats' interest in climate.

Staff progressively becoming more interested in climate may select into exposure to climate-attuned colleagues and subsequently exhibit greater focus on climate. To address this, we control for the count of a bureaucrat's prior climate recommendations, as noted above. We see confounding along these lines as unlikely due to the nature of team construction at the IMF. While climate is of growing interest at the Fund, managers in charge of assignments remain principally attentive to non-climate issues and thus are unlikely to make staff decisions based on a consideration of individual bureaucrats' climate preferences (Clark and Zucker 2024); until the 2020s, after the sample period, climate was not a key factor in the design of Article IV missions (Kentikelenis and Stubbs 2025). We estimate that bureaucrats' prior attentiveness to climate explains less than 2% of variation in subsequent exposure to climate-attuned colleagues.³⁰

Exposure to climate-attuned colleagues in one period may predict exposure to climate-attuned colleagues in subsequent periods. This is plausible if managers repeatedly seek to group climate-attentive bureaucrats together. Here we note that our control set includes the proportion of colleagues on a given mission team who themselves are attentive to climate (i.e., previously issued a climate-related policy recommendation). Holding constant

³⁰Regression of the share of climate-attuned colleagues on a mission on the count of a bureaucrat's prior climate recommendations. Estimated by OLS with bureaucrat and year fixed effects; within- R^2 of 1.8%. Prior climate recommendations have no significant relationship with future exposure when controlling for the climate attentiveness of prior coworkers ($p = 0.37$, standard errors clustered by individual).

the composition of a bureaucrat's current team, and bracketing common shocks via year fixed effects, change in a bureaucrat's behavior between two team assignments should be attributable to conditions on that first team.

An additional concern may be that a host country's interest in climate policy, or the severity of climate change in a host country, confounds the relationship between professional networks and investments of effort in climate policy recommendations. Here we note that country-year fixed effects should capture time-variant differences in countries' vulnerability to climate change. In other models, we control for country-year counts of climate disasters to account for variation in the salience of climate change within countries.

Results

Estimation results in Table 2 support our theory. At baseline, climate-relevant recommendations tend to include more complex language than those concerning other topics. This gap widens for bureaucrats who previously worked alongside more climate-attuned staff. These estimates are precise and substantively meaningful. Results of Model 3 indicate that the growth in the average climate attentiveness of colleagues observed between 2015 and 2019 would increase the complexity of future climate recommendations by 27% of a standard deviation relative to non-climate recommendations. To illustrate this increase qualitatively, IMF staff in 2015 recommended that Vanuatu set up “a natural disaster contingency fund ... funded over the medium term from the budget and by development partners.” Increasing complexity by 27% of a standard deviation would yield the recommendation that Vanuatu, in the following year, “[invest] in adaptation (either infrastructure such as sea-walls, breakwater reefs, that can withstand stronger weather events; or through relocating key infrastructure and conducting disaster mapping) [... and ensure that] the maintenance of critical infrastructure is adequately funded and part of a medium-term expenditure framework (e.g., stable allocations for the infrastructure maintenance fund, and including donor

support).”

The magnitude and significance of this result are consistent across covariate specifications and robust to demanding fixed effect batteries. We further find consistent results when respecifying the outcome as median recommendation complexity (Appendix H), as well as when using our alternative measure of bureaucratic effort: the specificity of recommendations (Appendix I). Results of these models indicate that the growth in average climate attentiveness between 2015–19 would increase the relative specificity of climate recommendations by 32–35% of a standard deviation.

Dependent variable: Mean complexity (standardized)				
	(1)	(2)	(3)	(4)
Climate-relevant $\times$ past coworkers (ln)	0.086** (0.028)	0.086** (0.028)	0.098** (0.030)	0.116*** (0.030)
Climate-relevant recommendation	0.180*** (0.032)	0.193*** (0.032)	0.223*** (0.035)	0.212*** (0.037)
Past coworkers’ climate attentiveness (ln)	−0.028** (0.010)	−0.004 (0.014)	0.023 (0.034)	−0.021** (0.008)
N	4367	4367	4367	4367
Adj. R ²	0.064	0.067	0.048	0.356
Control: climate attentiveness		✓	✓	✓
Control: years in sample		✓		
Control: climate disasters		✓	✓	
Control: % coworkers climate attuned		✓	✓	
Fixed effect: year	✓	✓	✓	
Fixed effect: bureaucrat			✓	✓
Fixed effect: report (country-year)				✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 2: Regressions of the mean complexity of policy recommendations coauthored by a bureaucrat in a given year on the climate relevance of those recommendations (binary), interacted with the climate attentiveness of past colleagues. Models estimated by OLS with standard errors clustered by bureaucrat.

One concern is that reports that discuss climate may be systematically distinct from

those that do not. This is plausible if climate recommendations primarily target more climate-vulnerable countries, which tend to feature lower state capacity. If economic governance challenges tend to be more acute in those countries, or if those countries rely more on the IMF for policy advice than other countries, Fund staff may construct more detailed policy recommendations for those countries across issue areas. The inclusion of report fixed effects in Model 4 accounts for this alternative explanation. Estimation of this model indicates that within the same report, the complexity gap between climate and non-climate recommendations widens when contributing bureaucrats previously worked alongside more climate-attuned colleagues.

We theorize that this growth in complexity owes to staff learning from individual colleagues, either via a subtle socialization process or via a material calculation of the professional returns to climate policymaking. To evaluate this peer-to-peer account, we examine whether a bureaucrat’s climate policy recommendations are more similar to the climate recommendations issued by past coworkers than the climate recommendations issued by staff that the bureaucrat did *not* work with. To do so, we make use of sentence embeddings, converting the text of climate recommendations into vectors that can then be compared to assess their semantic similarity.³¹ We calculate the cosine similarities between climate recommendations authored by bureaucrat i in a given report and (a) climate recommendations authored by past teammates of bureaucrat i (excluding any prior reports where bureaucrat i was also a coauthor), and (b) climate recommendations authored by staff with whom the bureaucrat had *not* worked alongside.

We find that climate recommendations resemble those of past teammates more than those of other staff members. On average, the cosine similarity to teammates’ recommendations is 39% of a standard deviation greater than the similarity to non-teammate

³¹We do so via the `all-mpnet-base-v2` sentence-transformers model (Song et al. 2020), widely used for semantic similarity tasks and regarded as the “best quality ... general purpose [sentence embedding] model” (SBERT 2026, perma.cc/EBT6-FYX8). Nearly all observations fall within the model’s token limit.

recommendations (Appendix J plots the distributions of cosine similarity).³² This result is important for two reasons. First, it supports our claim that interactions between individuals shift bureaucrat behavior; the manner by which staff discuss climate reflects the specific manner by which their peers discussed the issue. Second, it alleviates concerns around biased selection into exposure to climate-attuned colleagues. If a bureaucrat's prior interest in climate confounded the relationship between colleagues and subsequent recommendations (i.e., if interactions with those colleagues had no effect), we would not expect greater semantic similarity between that bureaucrat's recommendations and those specifically of their teammates. The similarities we uncover suggest that peer-to-peer interactions meaningfully shape bureaucrats' future climate work.

These results collectively indicate that bureaucrats learn from individual colleagues within the institution. While incentives imposed by principals and executives within the IMF may additionally spur behavioral change, we find that at any point in time, bureaucrats who previously worked alongside more climate-attuned colleagues produce more complex climate policy recommendations. Bureaucrats thus do not allocate effort across issues in isolation, but rather as a function of the types of colleagues with whom they engage.

LEARNING ACROSS RANKS

IMF surveillance work is organized around small, hierarchical mission teams that include a mission chief, a resident representative, and several junior staff. Mission chiefs lead these teams, coordinate interactions with country authorities,³³ and are ultimately responsible for the content of Article IV reports. Junior staff contribute analysis and draft policy rec-

³²Intercept-only OLS with standard errors clustered by bureaucrat and report; average difference = 0.045 cosine units ($p < 0.001$) with 235 bureaucrat-report observations. Standard deviation in cosine similarity = 0.116.

³³Mission chiefs and resident representatives serve as the primary points of contact with external actors, while junior staff are more often involved in analytical and drafting roles. By examining how prior exposure to climate-attuned colleagues affects future outputs, holding constant current team composition and country-level conditions, we isolate intra-bureaucratic mechanisms from contemporaneous external influences.

ommendations, which are reviewed and revised under the supervision of senior colleagues. This structure facilitates repeated, close interactions between junior and senior staff and gives senior personnel substantial influence over the evaluation of individual contributions.

Professional advancement within the IMF follows a meritocratic model in which performance evaluations, supervisory feedback, and demonstrated policy competence guide promotions and assignments. Because these evaluations are shaped in part by senior colleagues, junior staff have incentives to align their efforts with the priorities and expectations of those superiors. This is particularly relevant in emergent issue areas, where formal guidance is limited or inconsistent, and where staff preferences may vary widely. Here we evaluate whether bureaucrats learn differently from colleagues at different levels of the organizational hierarchy, as we theorize.

In the following tests, we thus disaggregate staff and their prior colleagues by rank: senior colleagues (mission chiefs or resident representatives while a focal bureaucrat was a junior staffer), junior colleagues (junior staffers while a mission chief or resident representative), or colleagues of equal rank (other junior staffers while a junior staffer, or mission chiefs or resident representatives while a fellow mission chief or resident representative). We estimate models of the same form as in Table 2.

Results

Estimates in Table 3 indicate that staff are most clearly responsive to the climate attentiveness of superiors. Model 2 suggests that the 2015–19 average increase in climate attentiveness, if present among prior senior coworkers, would increase the relative complexity of subsequent climate recommendations by 52% of a standard deviation. Exposure to climate-attuned colleagues of equal rank, as well as climate-attuned junior coworkers, lacks any significant association with future recommendation complexity; the point estimates are also notably smaller than those for senior coworkers. These results are consistent

when redefining the outcome as median complexity (Appendix K). With the alternative outcome of recommendation specificity, we find responsiveness to both senior and equal-rank colleagues, and a continued lack of clear responsiveness to junior coworkers (Appendix L).

Dependent variable: Mean complexity (standardized)			
	(1)	(2)	(3)
Climate-relevant $\times$ senior coworkers (ln)	0.176*** (0.047)	0.201*** (0.049)	0.224*** (0.045)
Climate-relevant $\times$ equal-rank coworkers (ln)	0.034 (0.036)	0.050 (0.037)	0.055 (0.037)
Climate-relevant $\times$ junior coworkers (ln)	0.044 (0.055)	0.031 (0.059)	0.072 (0.052)
Climate-relevant recommendation	0.182*** (0.032)	0.224*** (0.034)	0.214*** (0.036)
Past senior coworkers' climate attentiveness (ln)	0.022 (0.023)	−0.038 (0.071)	−0.043*** (0.011)
Past equal-rank coworkers' climate attentiveness (ln)	−0.042*** (0.012)	0.051 (0.037)	−0.011 (0.009)
Past junior coworkers' climate attentiveness (ln)	−0.024+ (0.015)	−0.020 (0.038)	−0.014 (0.012)
N	4367	4367	4367
Adj. R ²	0.047	0.052	0.365
Control: climate attentiveness		✓	✓
Control: climate disasters		✓	
Control: % coworkers climate attuned		✓	
Fixed effect: year	✓	✓	
Fixed effect: bureaucrat		✓	✓
Fixed effect: report (country-year)			✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 3: Regressions of the mean complexity of policy recommendations coauthored by a bureaucrat in a given year on the climate relevance of those recommendations (binary), interacted with the climate attentiveness of past coworkers (disaggregated hierarchically in relation to given bureaucrat). Models estimated by OLS with standard errors clustered by bureaucrat.

These results suggest that senior bureaucrats play an important role in shaping the pri-

orities and incentives of junior colleagues. On small mission teams, senior staff influence both the flow of information and the salience of particular issues, making climate more likely to be discussed and incorporated when it is of interest to the team lead. At the same time, the climate attitudes of senior colleagues plausibly affect junior staff's assessment of career incentives. Working under a climate-attuned mission chief may signal that reallocating effort toward climate is professionally valued and likely to be rewarded, given superiors' influence over evaluations and advancement. Finally, socialization pressures may reinforce these effects: junior staff are especially likely to emulate respected and successful colleagues, so exposure to climate-attuned senior staff can shape behavior through both career-related expectations and emulation. Significant results for interactions with equal-rank staff in supplementary analyses similarly point to a socialization process, insofar as those staff have less sway over career prospects (Appendix L).

In an additional test, we probe an implication of this finding. If bureaucrats invest effort in climate as a career-motivated signal to superiors, or because they have been socialized into placing greater value on climate work, we should see discussions of climate becoming more prominent in Article IV reports. To test this, we leverage the hierarchical structure of some recommendations in Article IV reports. Roughly 13% of recommendations in our sample include at least one *sub*-recommendation: additional policy advice nested under the primary recommendation. Nested recommendations are less focal and plausibly less visible to readers. We find that bureaucrats who previously worked alongside more climate-attuned senior colleagues are significantly more likely to discuss climate in a primary policy recommendation (i.e., less likely to nest it under recommendations concerning other issues). We find that exposure to climate-attuned colleagues of equal rank and junior colleagues has no significant relationship with the placement of climate policy recommendations (Appendix M), underscoring the unique importance of senior colleagues in shaping bureaucratic decision-making.

CONCLUSION

IOs manage wide ranges of policy challenges. Which issues do international bureaucrats prioritize in their work? Principals and institutional leaders often send inconsistent signals on the value of governing emergent policy challenges. We argue that the social nature of bureaucrats' work guides their decision-making in these cases. With original data on individual career paths and bureaucratic outputs at the IMF, we find that bureaucrats devote more effort to climate-related policy work after interacting with climate-attuned colleagues, producing more sophisticated climate advice relative to their own work in other domains. These effects are asymmetric across the bureaucratic hierarchy. Bureaucrats react strongly to the climate attentiveness of senior colleagues, but exhibit little responsiveness to the preferences of junior staff; we obtain mixed evidence of responsiveness to colleagues of equal rank. In sum, bureaucrats do not decide what to work on in isolation, nor do they mechanically follow formal mandates. Instead, they infer which issues merit sustained attention by observing the behavior of colleagues, especially those who occupy positions of authority within their organization.

Influential theories portray international organizations as rigid, adapting primarily to exogenous shocks or shifts in principal preferences (Krasner 1976; Wallander 2000; Colgan, Keohane, and Van de Graaf 2012). Generally, institutional change is attributed more to external forces than internal ones (Gerschewski 2021). This paper identifies an alternative pathway by which institutions reorient their work even in the absence of revised mandates or clear principal instruction. Routine personnel management practices, such as team assignments and rotations, allow new priorities to diffuse through bureaucratic hierarchies. Institutional change is not only imposed from the outside, but can emerge endogenously as bureaucrats evaluate and trade off between work tasks.

Our findings point to the value of studying bureaucrats in the context of the social en-

vironments in which they work. Prominent scholarship asserts that bureaucrats hold fixed preferences within institutions as a result of selection processes and self-reinforcing organizational cultures (Weaver 2008; Chwioroth 2015; Lang, Wellner, and Kentikelenis 2025). Studies of change in bureaucratic behavior often examine individual bureaucrats of a specific rank in isolation (Clark and Zucker 2024). Assessments of how bureaucrats affect institutional performance examine staff as individuals operating independently of their colleagues (Heinzel 2022). We show instead that internal social dynamics, particularly those structured by bureaucratic hierarchies, guide how organizations allocate effort, set agendas, and confront emerging policy challenges. Future research should therefore take seriously how interactions among bureaucrats shape policymaking, alongside the more familiar roles of principals, mandates, and external pressure.

There are at least two promising directions for work along these lines. First, we theorize that ideas spread within organizations partly due to the persuasive efforts of individual staff. One compelling line of research would involve characterizing the origins and strategies of such enterprising staff, possibly drawing on theories of norm entrepreneurship (Finnemore and Sikkink 1998). These bureaucrats may adopt distinct strategies for advancing their preferences when executives and principals are more or less ideologically aligned with them; when leadership is ambivalent, for example, staff may seek to assemble coalitions of colleagues to make their case, or find ways to more quietly promote their interests. Second, the career incentives we theorize are focused on within-institution professional advancement. An alternative possibility is that bureaucrats invest in certain issues, or seek to acquire certain skills, to make themselves more attractive to outside employers; observation of peers moving between institutions may change what bureaucrats prioritize. In these ways, and in the manner we describe in this paper, interactions between staff may meaningfully shape bureaucratic behavior and, in turn, the trajectories of their institutions.

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APPENDICES

A. Name Matching	2
B. Extraction of Policy Recommendations	3
C. Climate Keywords	4
D. LLM-Based Recommendation Classification	5
E. Keyword-Assisted Topic Models	6
F. LLM-Based Policy Categorization	7
G. Specificity Index Construction	8
H. Effort Reallocation: Median Complexity	9
I. Effort Reallocation: Specificity Outcome	10
J. Distribution of Cosine Similarity	11
K. Colleagues by Rank: Median Complexity	12
L. Colleagues by Rank: Specificity Outcome	13
M. Colleagues by Rank: Issue Nesting	14

A. NAME MATCHING

We track individual IMF bureaucrats over time using the author lists included in Article IV staff reports. We primarily link bureaucrats based on exact name matches. However, to account for slight differences in name spelling across reports, we also link a small number of bureaucrats based on the Levenshtein distance between two names, which records the minimum number of character deletions, insertions, and substitutions required to match two strings. This is a metric widely used for approximate string matching.

We link bureaucrats conservatively, only matching names that have a Levenshtein distance of one (i.e., just a single character adjustment) to minimize false positives. Prior to matching, we trim white space and remove prefixes from names (e.g., “Ms.”). For example, “Pierre-Paul” and “PierrePaul” have a Levenshtein distance of one. We exclude cases where one name has multiple matches meeting this condition. We additionally require that two authors’ initials match in order to proceed with linking. We conduct this procedure on 3.2 million name pairs (after excluding exact-match names), arriving at 50 approximate name matches.

B. EXTRACTION OF POLICY RECOMMENDATIONS

Research team members read each sampled Article IV report in full and manually extracted the text of individual policy recommendations. Recommendations were generally delivered in consistent format across reports. Figure B1 is a page from Cambodia's 2018 report. Highlighted are example policy recommendations that were extracted.

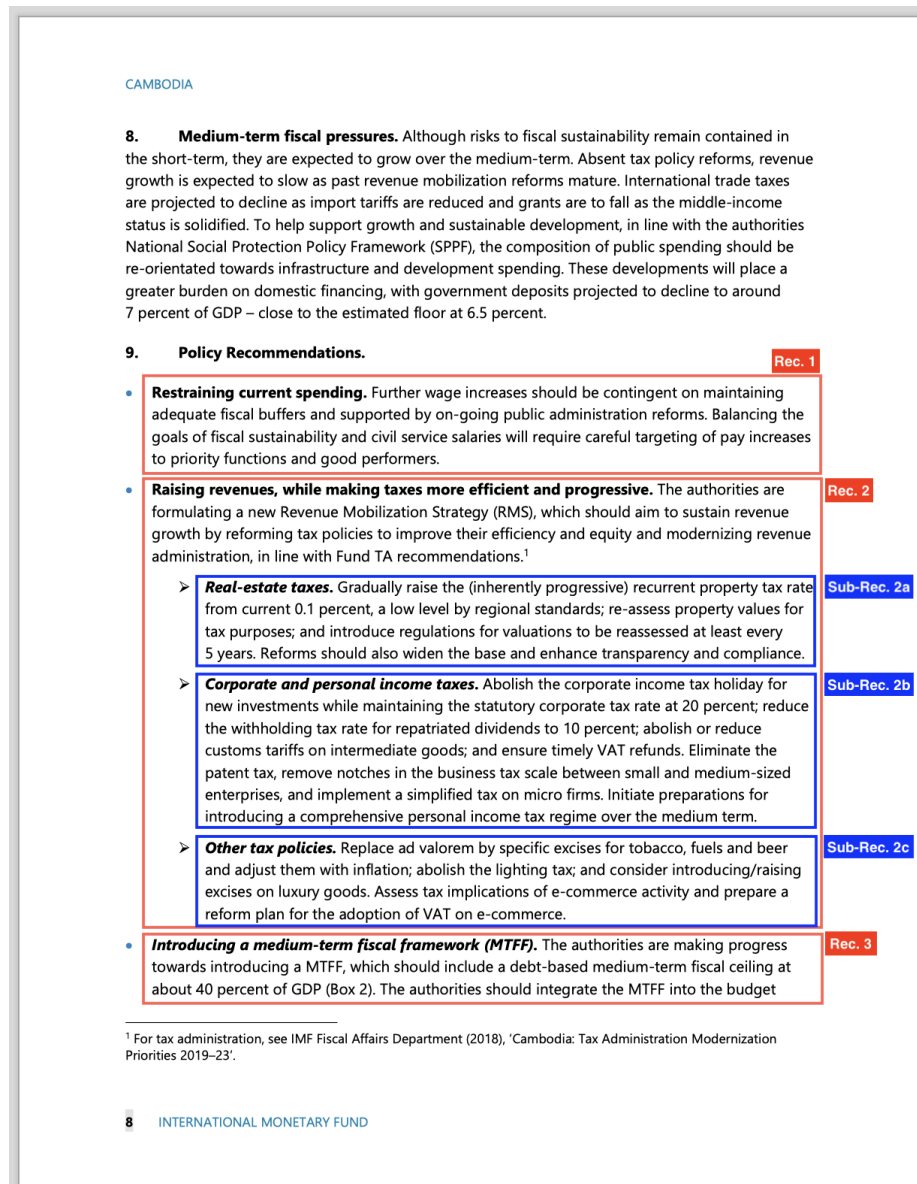


Figure B1: Page from Cambodia's 2018 Article IV report. Policy recommendations highlighted. Analyses of advice sophistication separate out sub-recommendations from parent recommendations (e.g., sub-recommendations 2a–c in the above would be considered separate from recommendation 2).

C. CLIMATE KEYWORDS

We conducted keyword searches of each extracted policy recommendation to identify those related to climate. Recommendations are coded as climate-relevant if they include at least one of the following terms:

- “adaptation”
- “bali action plan”
- “bali roadmap”
- “cap and trade”
- “carbon” (“hydrocarbon” cases excluded; recommendations only including the term “hydrocarbon” rarely focus on climate)
- “clean development mechanism”
- “climate change”
- “climate finance”
- “climate politics”
- “conference of the parties”
- “disaster risk”
- “disaster hazard”
- “emissions trading scheme”
- “framework convention on climate change”
- “ghg”
- “global average temperature”
- “global environmental facility”
- “global warming”
- “green climate fund”
- “greenhouse effect”
- “greenhouse gas”
- “environmental politic”
- “intergovernmental panel on climate change”
- “ipcc”
- “kyoto protocol”
- “mitigation”
- “nationally determined contribution”
- “natural disaster”
- “paris accord”
- “paris agreement”
- “renewables”
- “renewable energy”
- “unfccc”

This keyword search-based approach follows, among others, Clark and Zucker (2024).

D. LLM-BASED RECOMMENDATION CLASSIFICATION

We primarily identify climate-related recommendations via a keyword search (Appendix C). To validate this coding, we use the gpt-5-mini large language model (22 December 2025) to separately code recommendations extracted from Article IV reports. We indicate to the model that it should act as a “helpful, careful research assistant.” We prompt the model as follows:

The following is a policy recommendation issued by the IMF to a member state. Carefully determine whether the recommendation concerns climate change, decarbonization, or green/renewable energy (a recommendation about fossil fuels without reference to climate, decarbonization, or climate change mitigation does not count). Recommendations about natural disasters should be considered as being about climate change. If it is about climate change or decarbonization, return the number 1. If it is not, return the number 0. Include nothing else in your output. Here is the recommendation: [recommendation text]

Of the 234 identified via the keyword search, 204 (87%) were likewise classified as climate-relevant by the LLM.

E. KEYWORD-ASSISTED TOPIC MODELS

We estimate a keyword-assisted topic model (keyATM) to classify recommendations as adaptation- or mitigation-related. Proposed by Eshima, Imai, and Sasaki (2024), keyATM is a semisupervised approach to quantitative text analysis that permits more precise and guided extraction of themes from a text corpus than traditional automated topic models. To differentiate climate policy recommendations, we prepare and feed into keyATM sets of terms that tend to appear in either adaptation- or mitigation-focused recommendations (see below). Averaging across parameterizations, adaptation-related recommendations exceed mitigation-related recommendations in frequency by 33 percentage points (Figure E1).³⁴

- **Adaptation keywords:** "adapt", "adaptation", "resilience", "resilient", "coastal", "storms", "drought", "droughts", "cyclone", "cyclones", "hurricane", "hurricanes", "flood", "flooding", "temperature", "temperatures", "natural", "disaster", "disasters", "vulnerable", "vulnerability", "extreme", "weather"
- **Mitigation keywords:** "emission", "emissions", "energy-efficient", "pollution", "greenhouse", "renewable", "renewables", "clean", "solar", "green", "sustainability", "dirty", "fossil", "paris", "hydrocarbon", "hydrocarbons", "non-hydrocarbon", "oil", "gas", "coal", "petroleum"

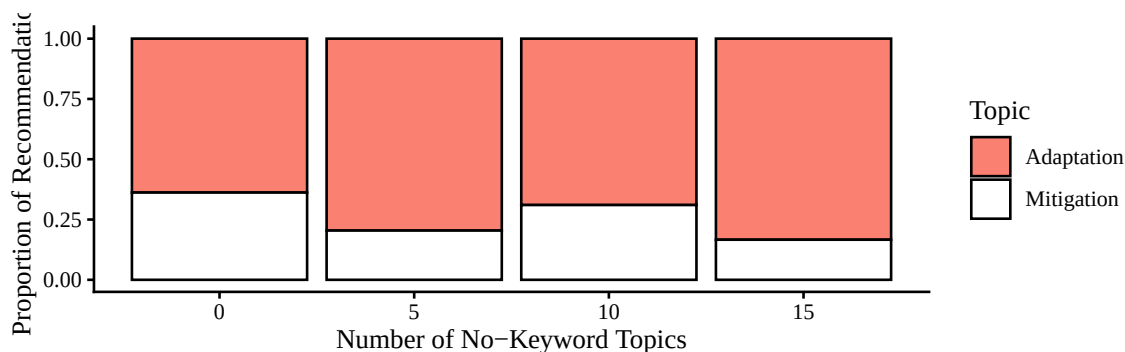


Figure E1: Proportions of climate policy recommendations that focus on adaptation or mitigation. Estimated for four no-keyword topic parameterizations, following Eshima, Imai, and Sasaki 2024.

³⁴keyATM allows researchers to instruct the model to generate “no-keyword topics” that identify themes distinct from the keyword-set topics. Following Eshima, Imai, and Sasaki 2024, we fit the model with zero, five, ten, and fifteen no-keyword topics to assess robustness (plotted in Figure E1).

F. LLM-BASED POLICY CATEGORIZATION

We categorize the policies discussed in Article IV climate-related policy recommendations via the gpt-4o large language model (22 December 2025). We prompt the model as follows:

I am going to list some policy recommendations issued by the IMF to member states. I want you to identify the categories of policies discussed in each recommendation. These categories should be: natural disaster management (including general adaptation policies), fiscal policy (including debt management, deficits, and fiscal buffers), public sector employment and bureaucracies, insurance, tax policy (including carbon prices), labor policy, infrastructure investments, ease of doing business, monetary policy, welfare policies, financial sector risk, uncategorized mitigation/decarbonization. Be precise in your categorizations. You can list multiple categories for each recommendation, but do so only if it is necessary to capture the breadth of the recommendation. Return this in CSV form, with one column for the number of the recommendation, and then subsequent columns for each category you decide upon (1 = recommendation fits in category; 0 = it does not fit).

Upon completion of the LLM process, we then conducted manual checks of random recommendations to ensure appropriate categorization; no manual changes were made to the LLM output. Of the 234 climate-related recommendations identified, 233 were sorted into at least one category. Of these, 192 (82%) were sorted into one or two policy categories. Thirty recommendations were sorted into three categories, 10 into four, and one into five.

G. SPECIFICITY INDEX CONSTRUCTION

The specificity index comprises three components: counts of references to specific policy instruments, temporal benchmarks, and numerics in individual policy recommendations. We standardize each component and then average across them to create a z-score index.

- **Policy instruments.** To identify references to specific policy instruments, we count per recommendation the following terms: "fiscal rule", "spending cap", "deficit target", "debt anchor", "building code", "zoning", "tax", "taxes", "levy", "fee", "fees", "subsidy", "subsidies", "tariff", "quota", "cap", "ceiling", "limit", "ban", "mandate", "standard", "regulation", "permit", "permits", "license", "licensing", "auction", "insurance", "buffer", "reserve", "benchmark", "target".
- **Temporal benchmarks.** To identify references to specific time benchmarks, we count per recommendation the following terms: "short[-]term", "medium[-]term", "long[-]term", "by [year]", "in [year]", "within [x] year[s]", "annual[ly]", "monthly", "quarterly", "yearly".
- **Numeric density.** To measure numbers, we count per recommendation the number of numeric tokens (e.g., "1,000"), percentage expressions (e.g., "10%" or "20 percent"), and currency references (e.g., "\$" or "euro").

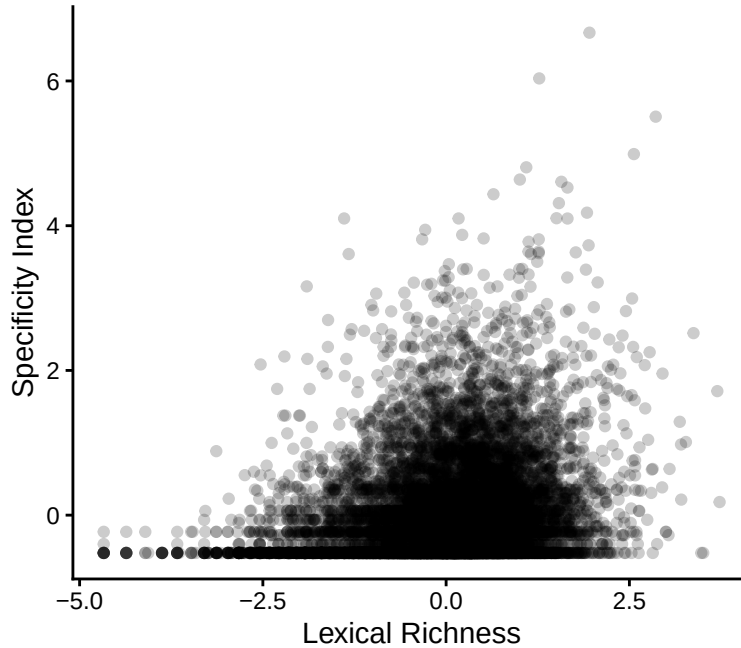


Figure G1: Recommendations plotted by lexical richness (primary measure of complexity) and the specificity index (secondary measure). Pearson’s correlation coefficient $\rho = 0.22$.

H. EFFORT REALLOCATION: MEDIAN COMPLEXITY

Dependent variable: Median complexity (standardized)				
	(1)	(2)	(3)	(4)
Climate-relevant $\times$ past coworkers (ln)	0.078** (0.030)	0.079** (0.030)	0.091** (0.031)	0.100** (0.031)
Climate-relevant recommendation	0.102*** (0.032)	0.114*** (0.032)	0.148*** (0.033)	0.148*** (0.035)
Past coworkers' climate attentiveness (ln)	-0.037*** (0.010)	-0.011 (0.014)	0.017 (0.035)	-0.018* (0.007)
N	4367	4367	4367	4367
Adj. R ²	0.044	0.047	0.042	0.354
Control: climate attentiveness		✓	✓	✓
Control: years in sample		✓		
Control: climate disasters		✓	✓	
Control: % coworkers climate attuned		✓	✓	
Fixed effect: year	✓	✓	✓	
Fixed effect: bureaucrat			✓	✓
Fixed effect: report (country-year)				✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table H1: Replication of Table 2, replacing the mean complexity outcome with median complexity.

I. EFFORT REALLOCATION: SPECIFICITY OUTCOME

Dependent variable: Mean specificity				
	(1)	(2)	(3)	(4)
Climate-relevant $\times$ past coworkers (ln)	0.130*** (0.030)	0.131*** (0.029)	0.126*** (0.030)	0.138*** (0.032)
Climate-relevant recommendation	0.074** (0.027)	0.055* (0.027)	0.062* (0.028)	0.063* (0.028)
Past coworkers' climate attentiveness (ln)	-0.015** (0.005)	-0.040*** (0.010)	-0.006 (0.019)	-0.027*** (0.007)
N	4367	4367	4367	4367
Adj. R ²	0.067	0.087	-0.041	0.047
Control: climate attentiveness		✓	✓	✓
Control: years in sample		✓		
Control: climate disasters		✓	✓	
Control: % coworkers climate attuned		✓	✓	
Fixed effect: year	✓	✓	✓	
Fixed effect: bureaucrat			✓	✓
Fixed effect: report (country-year)				✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table I1: Replication of Table 2, replacing the mean complexity outcome with mean specificity.

J. DISTRIBUTION OF COSINE SIMILARITY

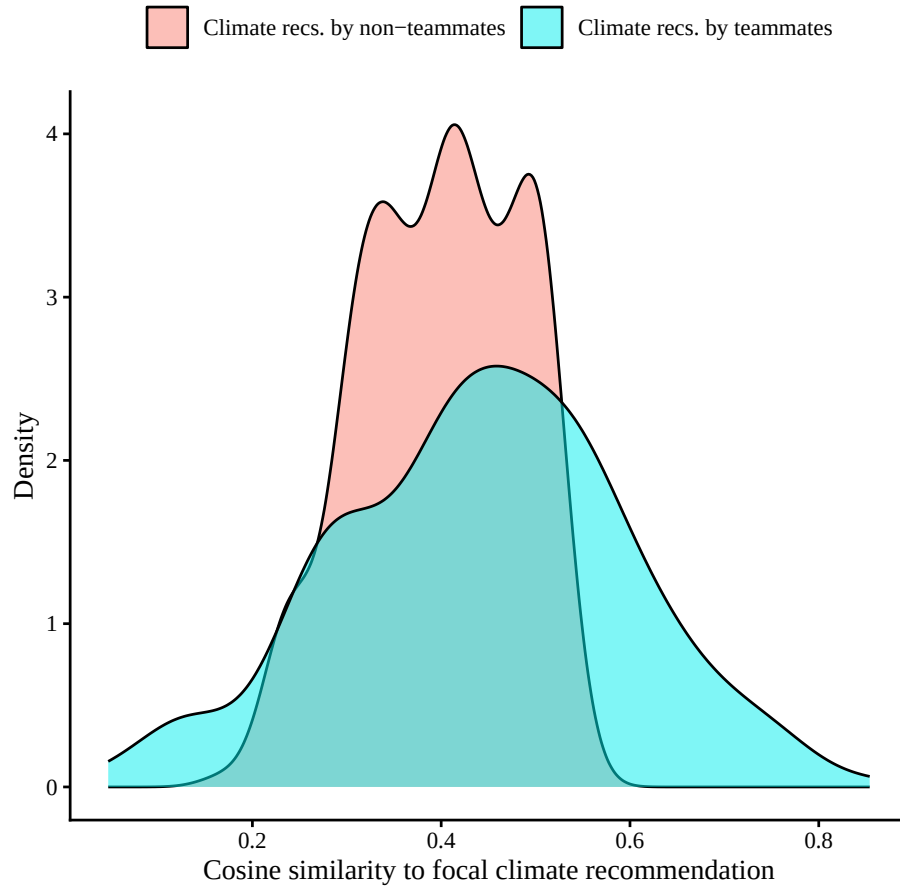


Figure J1: Distribution of cosine similarities between a bureaucrat’s climate recommendation and climate recommendations authored by (a) prior teammates of that bureaucrat and (b) other IMF staff, whom the bureaucrat did not previously work alongside.

K. COLLEAGUES BY RANK: MEDIAN COMPLEXITY

Dependent variable: Median complexity (standardized)			
	(1)	(2)	(3)
Climate-relevant $\times$ senior coworkers (ln)	0.197*** (0.051)	0.219*** (0.052)	0.237*** (0.050)
Climate-relevant $\times$ equal-rank coworkers (ln)	0.026 (0.038)	0.033 (0.039)	0.030 (0.038)
Climate-relevant $\times$ junior coworkers (ln)	0.039 (0.058)	0.028 (0.061)	0.062 (0.053)
Climate-relevant recommendation	0.102** (0.031)	0.152*** (0.033)	0.149*** (0.034)
Past senior coworkers' climate attentiveness (ln)	-0.036+ (0.020)	-0.083 (0.070)	-0.046*** (0.013)
Past equal-rank coworkers' climate attentiveness (ln)	-0.038*** (0.011)	0.045 (0.038)	-0.005 (0.009)
Past junior coworkers' climate attentiveness (ln)	-0.026+ (0.015)	-0.023 (0.039)	-0.011 (0.012)
N	4367	4367	4367
Adj. R ²	0.047	0.048	0.363
Control: climate attentiveness		✓	✓
Control: climate disasters		✓	
Control: % coworkers climate attuned		✓	
Fixed effect: year	✓	✓	
Fixed effect: bureaucrat		✓	✓
Fixed effect: report (country-year)			✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table K1: Replication of Table 3, replacing the mean complexity outcome with median complexity.

L. COLLEAGUES BY RANK: SPECIFICITY OUTCOME

Dependent variable: Mean specificity			
	(1)	(2)	(3)
Climate-relevant $\times$ senior coworkers (ln)	0.160** (0.052)	0.147** (0.052)	0.170** (0.057)
Climate-relevant $\times$ equal-rank coworkers (ln)	0.104** (0.037)	0.107** (0.036)	0.108** (0.037)
Climate-relevant $\times$ junior coworkers (ln)	0.078 (0.062)	0.061 (0.059)	0.081 (0.058)
Climate-relevant recommendation	0.075** (0.027)	0.062* (0.027)	0.065* (0.028)
Past senior coworkers' climate attentiveness (ln)	-0.011 (0.009)	0.072* (0.032)	-0.030* (0.014)
Past equal-rank coworkers' climate attentiveness (ln)	-0.016** (0.006)	-0.004 (0.019)	-0.024** (0.009)
Past junior coworkers' climate attentiveness (ln)	-0.007 (0.008)	-0.032 (0.028)	-0.018 (0.014)
N	4367	4367	4367
Adj. R ²	0.071	-0.032	0.053
Control: climate attentiveness		✓	✓
Control: climate disasters		✓	
Control: % coworkers climate attuned		✓	
Fixed effect: year	✓	✓	
Fixed effect: bureaucrat		✓	✓
Fixed effect: report (country-year)			✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table L1: Replication of Table 3, replacing the mean complexity outcome with mean specificity.

M. COLLEAGUES BY RANK: ISSUE NESTING

Dependent variable: Pr(non-nested recommendation = 1)			
	(1)	(2)	(3)
Climate-relevant $\times$ senior coworkers (ln)	0.070** (0.022)	0.043+ (0.025)	0.054* (0.025)
Climate-relevant $\times$ equal-rank coworkers (ln)	-0.029 (0.019)	-0.034+ (0.020)	-0.034+ (0.019)
Climate-relevant $\times$ junior coworkers (ln)	0.004 (0.027)	0.006 (0.029)	0.007 (0.027)
Climate-relevant recommendation	-0.008 (0.016)	-0.010 (0.015)	-0.020 (0.015)
Past senior coworkers' climate attentiveness (ln)	0.002 (0.014)	-0.074* (0.033)	-0.001 (0.001)
Past equal-rank coworkers' climate attentiveness (ln)	0.006 (0.008)	0.001 (0.017)	0.001+ (0.001)
Past junior coworkers' climate attentiveness (ln)	0.001 (0.011)	0.005 (0.019)	0.000 (0.001)
N	73577	73577	73577
Adj. R ²	0.018	0.127	0.178
Control: climate attentiveness		✓	✓
Control: climate disasters		✓	
Control: % coworkers climate attuned		✓	
Fixed effect: year	✓	✓	
Fixed effect: bureaucrat		✓	✓
Fixed effect: report (country-year)			✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table M1: Regressions of whether a given recommendation was not nested under another recommendation on the climate relevance of that recommendation, interacted with the climate attentiveness of past coworkers by rank. Models estimated by OLS with standard errors clustered by bureaucrat.